

# SOLVING THE EINSTEIN CONSTRAINT EQUATION ON CLOSED MANIFOLDS

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**ABSTRACT.** The initial data required to solve the Einstein Field Equations require certain restrictions and are not freely determinable. We investigate an approach from J. Isenberg [9] which utilises the space of constant mean curvature (CMC) solutions to the Einstein Constraint Equations on a closed manifold. The set of solutions are derived using a conformal formulation and are evaluated for their uniqueness.

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## 1. Background

Einstein's theory of general relativity models the geometry of space-time as a four-dimensional manifold. Finding general solutions to the Einstein Field Equations which describe this geometry is not particularly simple and involves a complicated system of non-linear equations [15, 16]. Moreover, the initial data to the Field Equations cannot be constructed freely but requires a set of constraint equations. The *Einstein Constraint Equations* allow for specific solutions and require making assumptions regarding the restrictions on the mean curvature, vacuum states, an overall separability of the partial differential equations (PDEs) involved. The mathematical description of this physical model as described by Isenberg [9] is as follows.

Let  $(\widetilde{\mathcal{M}}^4, \gamma)$  be a Lorentzian manifold of spatial dimension 3 and time dimension 1. Then the famous *Einstein Field Equations (EFE)* can be written as

$$\text{Ric}_\gamma - \frac{1}{2}R_\gamma\gamma + \Lambda\gamma = \kappa T.$$

Here  $\text{Ric}_\gamma$  is the Ricci curvature of the metric tensor  $\gamma$  with scalar curvature  $R_\gamma$ , and constant  $\kappa = 8\pi G/c^2$ ,  $T$  is the stress-energy tensor and  $\Lambda \in \mathbb{R}$  is known as the ‘cosmological constant’.

On a spacelike hyper-surface  $\mathcal{M}^3$  in  $\widetilde{\mathcal{M}}^4$ , a Riemannian metric tensor  $g_{ij}$  determines the intrinsic geometry of the manifold  $\mathcal{M}^3$ . The extrinsic geometry of  $\mathcal{M}^3$ , that is the way  $\mathcal{M}^3$  is curved within  $\widetilde{\mathcal{M}}^4$ , is determined by the second fundamental form  $K^{mn}$  (symmetric (2,0)-tensor). Considering a vacuum restriction to the EFE where  $\Lambda = 0$  and  $T \equiv 0$ , one can construct the *Einstein Constraint Equations* (ECE) such that for the set  $(g, K)$ :

$$R_g - \|K\|_g^2 + (\text{tr}_g K)^2 = 0 \quad (1a)$$

$$\nabla_i K_j^i - \nabla_j (\text{tr}_g K) = 0 \quad (1b)$$

to find solutions for the Einstein Field Equations on  $\mathcal{M}^3 \times \mathbb{R}$ . Here  $\nabla_g$  is the covariant derivative corresponding to  $g$ ,  $R_g$  scalar curvature, and  $\text{tr}_g K := g_{ij}K^{ij}$ .

Generally, the set of solutions  $(g, K)$  which is necessary to solve the EFE is not trivial to find. In this paper we consider an approach suggested by Isenberg [9] with constant mean curvature solutions on a closed (compact with no boundary) manifold.

**Aim.** We investigate the possible solutions to the Einstein Constraint Equations on a closed manifold under suitable assumptions, and discuss their implications and limitations.

## 2. Approach

### 2.1. The Conformal Formulation.

A pair of Riemannian metrics  $g_{ij}$  and  $\lambda_{ij}$  are conformally related by a positive definite function  $\phi$  such that

$$g_{ij} = \phi^4 \lambda_{ij}. \quad (2)$$

Similarly the corresponding scalar curvature terms  $R_{(g)}$  and  $R_{(\lambda)}$  are related as

$$R_{(g)} = \phi^{-5} \left( -8 \nabla_{(\lambda)}^2 \phi + R_{(\lambda)} \phi \right) \quad (3)$$

where  $\nabla^2$  is the Laplacian operator compatible with its denoted metric.

In order to relate this back to the Einstein Constraint Equation, a decomposition of a trace free symmetric tensor  $\mu^{ij}$  is required such that

$$\nabla_{(g)}^i (\phi^{-10} \mu^{ij}) = \phi^{-10} \left( \nabla_{(\lambda)}^i \mu^{ij} \right) \quad (4)$$

This reformulation developed by Lichnerowicz, Choquet-Bruhat and York [4] results in the decomposed form of the initial data  $(g, K)$

$$g_{ij} = \phi^4 \lambda_{ij} \quad (5a)$$

$$K^{mn} = \phi^{-10} (\sigma^{mn} + L W^{mn}) + \frac{1}{3} \phi^{-4} \lambda^{mn} \tau \quad (5b)$$

where  $\sigma^{mn}$  is a symmetric, trace-free ( $g_{ij}\sigma^{ij} = 0$ ), divergence-free ( $\nabla_{(\lambda)}^i \sigma^{ij} = 0$ ) tensor,  $\tau$  is a scalar field,

$L$  is a conformal killing factor,  $W^a$  is a vector field and together

$$L W_{ij} = \nabla_{(\lambda)}^i W_j + \nabla_{(\lambda)}^j W_i - \frac{2}{3} \lambda_{ij} \nabla_{(\lambda)}^m W_m.$$

The decomposition for  $(g, K)$  satisfy the ECE if and only if the set  $(\lambda, \sigma, \tau; \phi, W)$  satisfy

$$\nabla_{(\lambda)}^2 \phi = \frac{1}{8} R_{(\lambda)} \phi - \frac{1}{8} (\sigma^{ij} + LW^{ij}) (\sigma_{ij} + LW_{ij}) \sigma^{-7} + \frac{1}{12} \tau^2 \phi^5 \quad (6a)$$

$$\nabla_{(\lambda)}^i (LW)_m^i = \frac{2}{3} \phi^6 \nabla_m \tau \quad (6b)$$

Crucially, solutions for  $(\lambda, \sigma, \tau)$  from equations (6a) and (6b) lead to solutions for the Einstein Constraint Equations. With further simplification of  $\tau = \text{const}$ , equations (6b) as well as  $LW_{ij}$  vanish, leading to the useful ‘Lichnerowicz’ equation,

$$\nabla_{(\lambda)}^2 \phi = \frac{1}{8} R_{(\lambda)} \phi - \frac{1}{8} (\sigma^{ij} \sigma_{ij}) \phi^{-7} + \frac{1}{12} \tau^2 \phi^5. \quad (7)$$

This crucial simplification of the *constant mean curvature* arrives from taking the trace of equation (6b)

$$\tau := \text{tr}_g K = \text{const} \quad (8)$$

and can geometrically be interpreted as the closed three-dimensional manifold ‘bending’ in the same way at every point<sup>1</sup>.

The problem of solving the ECE is hence reduced to finding  $\phi > 0$  and so by finding solution sets  $(\lambda, \sigma, \tau; \phi)$  for equation (7), the sets of suitable initial data  $(g, K)$  is determined and the solutions for the Einstein Constraint Equations are found.

## 2.2. Maximum Principles.

Different forms of the Maximum Principle exist and are used to solve elliptic partial differential equations. Extension of the proofs are stated by Protter [13] and we state a subset of the principles relevant to our proofs.

*Format 1.* On a closed manifold, if  $\phi$  satisfies either  $\nabla^2 \phi \geq 0$ ,  $\nabla^2 \phi \leq 0$ , or  $\nabla^2 \phi = 0$ , then  $\phi$  must be constant. Therefore there is no solution to

$$\nabla^2 \phi = F(x, \phi)$$

unless  $F(x, \phi) = 0$ .

*Format 2.* On a closed manifold, let  $f : \mathcal{M}^3 \rightarrow \mathbb{R}$  be a positive function. If  $\phi$  satisfies

$$\nabla^2 \phi \leq f\phi \quad (9)$$

then  $\phi(x) \geq 0$  (or  $\phi(x) > 0$  when equation (9) is *not* identically zero).

Similarly, if  $\phi$  satisfies

$$\nabla^2 \phi \geq f\phi \quad (10)$$

then  $\phi(x) \leq 0$  (or  $\phi(x) < 0$  when equation (10) is *not* identically zero).

*Format 3.* On an open manifold with boundary  $\partial\mathcal{M}^3$ , let  $\varepsilon > 0$  be a constant, and  $\varrho : \partial\mathcal{M}^3 \rightarrow \mathbb{R}$  be a positive function. If on  $\mathcal{M}^3$ ,  $\phi$  satisfies

$$\nabla^2 \phi = \varepsilon \phi, \quad (11)$$

and  $\phi = \varrho$  on  $\partial\mathcal{M}^3$ , then  $\phi > 0$ .

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<sup>1</sup>In the two-dimensional case, one can imagine a connected, closed-surface sphere  $S^2 \subset \mathbb{R}^3$  which has a plane of symmetry in every direction to withhold *constant* mean curvature.

### 2.3. The Yamabe Classification.

In order to reduce the variables in the Lichnerowicz equation (7), one can fix  $\frac{R}{(\lambda)}$  which is possible through the solution to the Yamabe Problem.

The Yamabe Problem is concerned whether conformal solutions such as equation (2) with constant mean curvature exist or not. The resolution, as widely studied by Aubin [1], and Schoen [14] depends on finding  $\theta > 0$  solution for the equation

$$\nabla_{(\lambda)}^2 \theta = \frac{1}{8} \frac{R}{(\lambda)} (\theta - \theta^5) \quad (12)$$

where  $\frac{R}{(\lambda)}$  is constant. Hence, given  $\lambda_{ij}$  a Riemannian metric on a closed manifold  $\mathcal{M}^3$ , there exists a positive definite function  $\theta$  on  $\mathcal{M}^3$  such that  $\frac{R}{(g)}$  is constant with  $g$  defined from equation (2). Although the complete proof of the theorem has an extensive process, the related proposition is noted here.

**2.3.1. The Classification Proposition.** Using Formats 1 and 2 of the Maximum Principle, it is understood that the conformal transformation to  $\lambda_{ij}$  may only yield a positive, negative, or zero scalar curvature. (i.e.  $\frac{R}{(\lambda)} = -1, 0, +1$  due to normalisation). It also follows that the conformal transformation to a metric with one of the three scalar curvature values cannot obtain any other constant scalar curvature values. This leads to the following three ‘Yamabe Classes’ with their corresponding metrics:

$$\mathcal{Y}^-(\mathcal{M}^3) = \{\text{smooth } \lambda_{ij} \mid \frac{R}{(g)} = -1\}$$

$$\mathcal{Y}^0(\mathcal{M}^3) = \{\text{smooth } \lambda_{ij} \mid \frac{R}{(g)} = 0\}$$

$$\mathcal{Y}^+(\mathcal{M}^3) = \{\text{smooth } \lambda_{ij} \mid \frac{R}{(g)} = +1\}$$

A useful corollary to this classification is that although for certain closed manifolds  $\mathcal{Y}^+$  or  $\mathcal{Y}^0$  can be empty,  $\mathcal{Y}^-$  is never empty and  $\mathcal{Y}^0$  is empty only if  $\mathcal{Y}^+$  is empty. It is also important to note that for cases where  $\sigma^2 \equiv 0$  in some regions  $\hat{\mathcal{M}}^3$  of  $\mathcal{M}^3$  and given  $\lambda_{ij} \in \mathcal{Y}^+$  or  $\mathcal{Y}^0$ , suitable conformal transformations with  $\frac{R}{(g)} < 0$  can be made. The following proposition by O’Murchadha et al. [12] addresses the previous statement.

**Proposition 1.**  $\mathcal{M}^3$  is a closed manifold with Riemannian metric  $\lambda_{ij}$  of some Yamabe Class.  $\hat{\mathcal{M}}^3$  is an open subset of  $\mathcal{M}^3$  with a regular boundary  $\partial\hat{\mathcal{M}}^3$ , and  $\mathcal{M}^3 - (\hat{\mathcal{M}}^3 \cup \partial\hat{\mathcal{M}}^3)$  is also open and non-empty in  $\mathcal{M}^3$ . Then there exists a positive definite function  $\theta$  on  $\mathcal{M}^3$  such that on  $\hat{\mathcal{M}}^3$

$$\frac{R}{(g)} < -\varepsilon < 0 \quad (13)$$

where  $\varepsilon$  is a positive constant.

*Proof:*  $\frac{R}{(\lambda)}$  is continuous as  $\lambda_{ij}$  is twice differentiable by definition. As  $\mathcal{M}^3$  is compact (without boundary), there must exist positive  $Q$  such that  $\frac{R}{(\lambda)} < Q$ . For the appropriate conformal transformation  $\tilde{\lambda}_{ij} := (2Q)^4 \lambda_{ij}$ , and using equation (3), it is true that  $\frac{R}{(\tilde{\lambda})} < \frac{1}{2}$ .

It is known that for the Dirichlet boundary value problem

$$8 \nabla_{(\tilde{\lambda})}^2 u = u \quad (\text{on } \mathcal{M}^3) \quad (14a)$$

$$u = 1 \quad (\text{on } \partial\mathcal{M}^3) \quad (14b)$$

a unique solution for  $u$  exists [6]. Furthermore  $u$  is positive definite due to the Maximum Principle (Format 3), and  $0 < u < m$  for positive  $m$  from the continuity argument. For a choice of  $C^3$  function  $\theta$  on  $\mathcal{M}^3$  where specifically  $\theta = u$  on  $\hat{\mathcal{M}}^3 \cup \partial\hat{\mathcal{M}}^3$ , one may derive the scalar curvature for  $g_{ij} := \theta^4 \tilde{\lambda}_{ij} = u^4 (2Q)^4 \lambda_{ij}$  on  $\hat{\mathcal{M}}^3$ . Note that the  $\theta$  exists as  $\mathcal{M}^3 - \hat{\mathcal{M}}^3 \cup \partial\hat{\mathcal{M}}^3$  is indeed open. Then

$$\begin{aligned} R_{(g)} &= u^{-4} R_{(\tilde{\lambda})} - 8u^{-5} \nabla_{(\tilde{\lambda})}^2 u \\ &= u^{-4} (R_{(\tilde{\lambda})} - 1) \end{aligned}$$

As  $R_{(\tilde{\lambda})} < \frac{1}{2}$ , then  $R_{(g)} < 0$  on  $\hat{\mathcal{M}}^3$ . Substituting  $u < m$  from before,

$$\begin{aligned} R_{(g)} &< u^{-4} \left(-\frac{1}{2}\right) \\ &< -\frac{1}{2} m^{-4}. \end{aligned}$$

Choosing  $\varepsilon := \frac{1}{2} m^{-4}$  yields the expected result from equation (13).

#### 2.4. Sub- and Super-Solution Method.

The Lichnerowicz equation is a quasi-linear partial differential equation which takes the form

$$\nabla^2 \psi = f(x, \psi) \tag{15}$$

for which the solutions can be found via the method of *sub/super-solutions* [2]. This standard method in PDE theory claims that if for a pair of functions  $\psi_-$  and  $\psi_+$ ,

$$\nabla^2 \psi_- \geq f(x, \psi_-(x)), \tag{16a}$$

$$\nabla^2 \psi_+ \leq f(x, \psi_+(x)), \tag{16b}$$

$$\psi_-(x) \leq \psi_+(x) \tag{16c}$$

then there exists a solution  $\psi$  for equation (15) with  $\psi_-(x) \leq \psi(x) \leq \psi_+(x)$ .

Using this method is crucial to finding solutions with initial data  $(\lambda, \sigma, \tau)$  in equation (7). The rigorous proof of the method for a closed manifold requires the use of Sobolev spaces  $H_k^p(M^n)$  and Hölder spaces  $C^{m,\alpha}(M^n)$ , the discussion of which is embedded in [2] and [9]. A brief strategy to the proof is as follows.

**Proposition 2:** For a closed manifold  $\mathcal{M}^3$ , let  $f : \mathcal{M}^3 \times \mathbb{R}^+ \rightarrow \mathbb{R}$  be a once-differentiable function. Assuming the sub and super solutions  $\phi_- : \mathcal{M}^3 \rightarrow \mathbb{R}^+$  and  $\phi_+ : \mathcal{M}^3 \rightarrow \mathbb{R}^+$  exist, then

$$(i) \quad \phi_-, \phi_+ \in H_2^p(\mathcal{M}^3) \quad \text{where } p > 3,$$

$$(ii) \quad 0 < \phi_-(x) \leq \phi_+(x) \quad \text{for all } x \in \mathcal{M}^3,$$

$$(iii) \quad \nabla^2 \phi_- \geq f(x, \phi_-), \quad \text{and}$$

$$(iv) \quad \nabla^2 \phi_+ \leq f(x, \phi_+).$$

Then there exists the solution  $\phi : \mathcal{M}^3 \rightarrow \mathbb{R}$  such that

$$(v) \quad \phi \in C^{2,\alpha}(\mathcal{M}^3) \quad \alpha \in (0, 1 - 3/p),$$

$$(vi) \quad \phi_-(x) \leq \phi(x) \leq \phi_+(x), \quad \text{and}$$

$$(vii) \quad \nabla^2 \phi = f(x, \phi).$$

*Sketch of Proof:* Naturally the proof involves using  $\phi_-$  and  $\phi_+$  to construct a sequence of functions  $\{\phi_n\}$  that converges to the solution  $\nabla^2 \phi = f(x, \phi)$ . The analytical proof is extensive but can generally be separated to distinguishable steps.

**Step 1.** Inequalities (16a-b) can be related to equation (vii) in an operator form where

$$\mathcal{L} := -\nabla^2 + \xi \quad (17a)$$

$$\mathcal{F}(x, s) := -f(x, s) + \xi s, \quad (17b)$$

such that

$$\nabla^2 \phi = f(x, \phi) \quad \leftrightarrow \quad \mathcal{L}\phi = \mathcal{F}(x, \phi) \quad (18a)$$

$$\nabla^2 \phi_+ \leq f(x, \phi_+) \quad \leftrightarrow \quad \mathcal{L}\phi_+ \geq \mathcal{F}(x, \phi_+) \quad (18b)$$

$$\nabla^2 \phi_- \geq f(x, \phi_-) \quad \leftrightarrow \quad \mathcal{L}\phi_- \leq \mathcal{F}(x, \phi_-) \quad (18c)$$

Given  $\xi > \frac{\partial}{\partial s} f(x, s)$  on  $D := \mathcal{M}^3 \times [\min_{\mathcal{M}^3} \phi_-, \max_{\mathcal{M}^3} \phi_+]$  which implies that  $\frac{\partial \mathcal{F}}{\partial s} \geq 0$ . As a consequence, if  $\mathcal{L}\phi \geq 0$ , then  $\phi$  satisfies  $\phi \geq 0$  on  $\mathcal{M}^3$ .

**Step 2.** Using the operators, define a sequence of functions  $\{\phi_n\}$  on  $\mathcal{M}^3$  such that

$$\mathcal{L}\phi_{n+1} = \mathcal{F}(x, \phi_n) \quad (19)$$

for which the monotonicity property can be proved such that

$$\phi_+(x) \geq \phi_1(x) \geq \dots \geq \phi_j(x) \geq \phi_{j+1}(x) \geq \dots \geq \phi_-(x) \quad (20)$$

**Step 3.** As  $\{\phi_n\}$  is bounded, the Arzela-Ascoli Theorem (see [11] chapter 7) proves that the sequence converges in  $C^0$ .

**Step 4.** Given  $\phi_\infty$  denotes the limit function and  $\phi_-(x) \leq \phi_\infty(x) \leq \phi_+(x)$  due to monotonicity, one can prove the following three properties which lead to  $\phi_\infty$  being a solution to equation (vii):

- (1)  $\phi_\infty \in H_2^p(\mathcal{M}^3) \subset C^{1,\alpha}(\mathcal{M}^3)$
- (2)  $\phi_\infty \in C^{2,\alpha}(\mathcal{M}^3)$
- (3)  $\phi_\infty$  is a solution of  $\nabla^2 \phi_\infty = f(x, \phi_\infty(x))$

This completes the sketch of proof of the sub/super-solution method purely for the *existence* of solutions.  $\square$

### 3. Results & Insights

Solutions for the set of data  $(\lambda, \sigma, \tau; \phi)$  in the Lichnerowicz equation (7) yields the results table below. The important factors in solving equation (7) are **i)** the inclusion of  $\lambda_{ij}$  in a relevant Yamabe Class, **ii)** whether  $\sigma^2 := \sigma_{ij}\sigma^{ij} = \|\sigma\|^2$  is or is not equivalent to 0 across  $\mathcal{M}^3$ , and **iii)** whether  $\tau$  is equal to zero or not.

| $R_\lambda$ | $\sigma^2 \equiv 0, \tau = 0$ | $\sigma^2 \equiv 0, \tau \neq 0$ | $\sigma^2 \not\equiv 0, \tau = 0$ | $\sigma^2 \not\equiv 0, \tau \neq 0$ |
|-------------|-------------------------------|----------------------------------|-----------------------------------|--------------------------------------|
| +1          | $\times$                      | $\times$                         | $\checkmark^*$                    | $\checkmark^*$                       |
| 0           | $\checkmark$                  | $\times$                         | $\times$                          | $\checkmark^*$                       |
| -1          | $\times$                      | $\checkmark^*$                   | $\times$                          | $\checkmark^*$                       |

TABLE 1. Results for the parametrisation  $(\lambda, \sigma, \tau; \phi)$ . \* indicates uniqueness of solutions

Note that in the class  $(\mathcal{Y}^0, \sigma^2 \equiv 0, \tau = 0)$  the solution is any constant (non-unique), whereas for other classes the existing solutions are unique. We proceed to show three examples for existence and uniqueness of solutions in the table above and expand on the general method on finding further solutions.

### 3.1. Existence.

*No Solution.* Consider the case for  $\mathcal{Y}^-(\mathcal{M}^3)$ ,  $\sigma^2 \equiv 0$ ,  $\tau = 0$ . After appropriate substitution into the Lichnerowicz equation (7),

$$\nabla_{(\lambda)}^2 \phi = -\frac{1}{8}\phi \quad (21)$$

By generalised Stokes' theorem,

$$\int_{\mathcal{M}^3} \nabla_{(\lambda)}^2 \phi = \int_{\partial \mathcal{M}^3} \nabla_{(\lambda)} \phi, \quad (22)$$

where  $\nabla_{(\lambda)}^2$  and  $\nabla_{(\lambda)}$  are the Laplacian operator and covariant derivatives of  $\lambda_{ij}$  respectively. From equation (21)

$$\int_{\mathcal{M}^3} \nabla_{(\lambda)}^2 \phi = \int_{\mathcal{M}^3} -\frac{1}{8}\phi < 0$$

As  $\phi > 0$ . But  $\partial \mathcal{M}^3 = \emptyset$  as  $\mathcal{M}^3$  is closed by definition with empty boundary. Hence  $\int_{\partial \mathcal{M}^3} \nabla_{(\lambda)} \phi = 0$ . As this is in contradiction to the equality from equation (22), then there are no solutions for  $\phi > 0$  for the Lichnerowicz equation.

For other no solution classes, the proofs become trivial via the Maximum Principle (Format 1) for which no positive definite  $\phi$  exists.

*Solution exists.* Consider the case for  $\mathcal{Y}^-(\mathcal{M}^3)$ ,  $\sigma^2 \not\equiv 0$ ,  $\tau \neq 0$ . From the Yamabe classification we choose  $R_{(\lambda)} = -1$  and the Lichnerowicz equation simplifies to

$$\nabla_{(\lambda)}^2 \phi = -\frac{1}{8}\phi - \frac{1}{8}\sigma^2\phi^{-7} + \frac{1}{12}\tau^2\phi^5 \quad (23)$$

With investigation from the  $\mathcal{Y}^-(\mathcal{M}^3)$ ,  $\sigma^2 \equiv 0$ ,  $\tau \neq 0$  case we pick the constant sub-solution

$$\phi_- = \sqrt[4]{\frac{3}{2\tau^2}}$$

Given a constant sub-solution,  $\nabla \phi_- = 0$ . By the sub/super-solution method we require that  $\nabla^2 \phi_- \geq f(x, \phi_-)$  where  $f$  is the RHS of equation (23). Hence evaluating equation (23) at  $\phi = \phi_-$  yields

$$\begin{aligned} -\frac{1}{8}\phi_- - \frac{1}{8}\sigma^2\phi_-^{-7} + \frac{1}{12}\tau^2\phi_-^5 &= -\frac{1}{8}\sigma^2\phi_-^{-8} \\ &\leq 0 \end{aligned} \quad (24)$$

and so the constant  $\phi_-$  is a sub-solution. For the super-solution we require that  $\nabla^2 \phi_+ \leq f(x, \phi_+)$ , and  $\phi_- \leq \phi_+$ . From the result of equation (24), it is sufficient to find  $\phi_+$  such that  $\nabla^2 \phi_+ \geq 0$ . Thus,

$$0 \geq -\frac{1}{8}\phi_+ - \frac{1}{8}\sigma^2\phi_+^{-7} + \frac{1}{12}\tau^2\phi_+^5$$

and rearranging gets

$$\phi_+^4 \geq \frac{3}{2\tau^2} (1 + \sigma^2\phi_+^{-8}) \quad (25)$$

Once again  $\phi_+ \geq \phi_- > 0$ , and one can see two possible results for  $\phi_+ \in (0, 1]$  or  $\phi_+ \in (1, \infty)$ . So the super-solution reduces to

$$\phi_+ = \max \left( 1, \sqrt[4]{\left(\frac{3}{2\tau^2}\right) (1 + \max \sigma^2)} \right) \quad (26)$$

which is a suitable solution and hence a solution  $\phi_- \leq \phi \leq \phi_+$  exists. The uniqueness of this solution will be discussed.

*Solution exists.* As another example, consider the case for  $\mathcal{Y}^0(\mathcal{M}^3)$ ,  $\sigma^2 \not\equiv 0$ ,  $\tau \neq 0$ . Substitution in the Lichnerowicz equation and choosing  $R_{(\lambda)} = 0$  from the given Yamabe classification yields

$$\nabla_{(\lambda)}^2 \phi = -\frac{1}{8}\sigma^2\phi^{-7} + \frac{1}{12}\tau^2\phi^5. \quad (27)$$

We start by the subclass where  $\sigma^2 \not\equiv 0$  *everywhere* on  $\mathcal{M}^3$ . By inspection from the previous case and solving, the constant sub- and super-solutions simply are

$$\begin{aligned} \phi_- &= \sqrt[12]{\left(\frac{3}{2\tau^2}\right) \min \sigma^2} \\ \phi_+ &= \sqrt[12]{\left(\frac{3}{2\tau^2}\right) \max \sigma^2} \end{aligned} \quad (28)$$

and a solution  $\phi$  can exist as before. For the subclass where  $\sigma^2$  can be zero on  $\mathcal{M}^3$ . Given now  $\phi_-$  can now be zero on  $\mathcal{M}^3$ , the results from proposition 2 cannot be used. Instead, we consider a point  $p \in \mathcal{M}^3$  where  $\sigma^2 \not\equiv 0$  with an open ball  $B$  (radius  $\varrho$ ) using metric  $\lambda_{ij}$  containing  $p$  such that  $\sigma^2 \geq \varrho > 0$  in  $B$ . Suppose  $\hat{\mathcal{M}}^3$  exists such that  $\mathcal{M}^3 = \hat{\mathcal{M}}^3 \cup B$  and so all zeros of  $\sigma^2$  exist inside  $\hat{\mathcal{M}}^3$  by definition. Using proposition 1 on  $\hat{\mathcal{M}}^3$ , there exists a conformal transformation  $\tilde{\lambda}_{ij} = \theta^4 \lambda_{ij}$  where  $R_{(\tilde{\lambda})} \leq -\varepsilon < 0$ . Similarly,  $\tilde{\sigma}^2 := \theta^{-12}\sigma^2$ .  $\theta$  is continuous on  $\mathcal{M}^3$  and maximises on the closure of  $B$ , hence  $\tilde{\sigma}^2 \geq \tilde{\varrho} > 0$  where  $\tilde{\varrho}$  is  $\varrho$  with  $\theta$  on the closure. Using the conformal data  $(\tilde{\lambda}, \tilde{\sigma}, \tau)$ , the sub- and super-solutions for constants  $\phi_+, \phi_-$  after rearranging the Lichnerowicz “inequality” are

$$\frac{2}{3}\tau^2\phi_+^5 \geq -R_{(\tilde{\lambda})}\phi_+ + \tilde{\sigma}^2\phi_+^{-7} \quad (29a)$$

$$\frac{2}{3}\tau^2\phi_-^5 \leq -R_{(\tilde{\lambda})}\phi_- + \tilde{\sigma}^2\phi_-^{-7} \quad (29b)$$

as  $\nabla_{(\tilde{\lambda})}^2 \phi_+ = \nabla_{(\tilde{\lambda})}^2 \phi_- = 0$  and now the scalar curvature is not constant. Equations (29a-b) are similar to equation (25), and after some rearrangement one obtains

$$\phi_+ = \max \left( 1, \sqrt[4]{\left(\frac{3}{2\tau^2}\right) (\max_{(\tilde{\lambda})} |R| + \max \tilde{\sigma}^2)} \right) \quad (30a)$$

$$\phi_- = \min \left( 1, \sqrt[8]{\tilde{\varrho} \left( \frac{1}{\max_{(\tilde{\lambda})} |R| + \frac{2}{3}\tau^2} \right)}, \sqrt[4]{\frac{2}{3} \frac{\kappa}{\tau^2}} \right). \quad (30b)$$

For the sub-solution, we notice that  $\tilde{\sigma}^2 \geq \tilde{\varrho} > 0$  on closure of  $B$  while recalling that  $\tilde{\sigma}^2$  can be zero on  $\hat{\mathcal{M}}^3$  and by proposition 1  $R_{(\tilde{\lambda})} \leq -\kappa < 0$  which results into another possible restriction for the sub-solution such that

$$\phi_- \leq \sqrt[4]{\frac{3}{2\tau^2} \kappa}$$

and equation (30b) is obtained. On this basis  $\phi_+ \geq \phi_- > 0$  and hence a solution  $\phi$  exists in between.

The solution for the  $(\mathcal{Y}^-, \sigma^2 \equiv 0, \tau \neq 0)$  class is  $\phi = \sqrt[4]{3/2\tau^2}$ . As shown in the last two examples, the other existence solutions follow the same path, using this answer as hypothesis while applying



proposition 2.

### 3.2. Uniqueness.

The only non-unique solution, as mentioned in the results, is the class for  $\mathcal{Y}^0(\mathcal{M}^3)$ ,  $\sigma^2 \equiv 0$ ,  $\tau = 0$  where the solution is trivial (flat and static space-time) and any constant as the Lichnerowicz equation takes the form  $\nabla_{(\lambda)}^2 \phi = 0$ . For the other two uniqueness proofs, we follow the results of the following lemma.

3.2.1. *Lemma.* For  $f : \mathcal{M}^3 \times \mathbb{R}^3 \rightarrow \mathbb{R}$  a once-differentiable function with

$$\frac{\partial f}{\partial s}(x, s) \geq 0 \quad (31)$$

for  $s \in I$  (possibly infinite) interval in  $\mathbb{R}^+$ . If  $\psi_1$  and  $\psi_2$  are solutions to the previously mentioned PDE,

$$\nabla^2 \psi = f(x, \psi(x)), \quad (32)$$

and if  $\psi_1$  and  $\psi_2$  exist within  $I$  for all  $x \in \mathcal{M}^3$ , then  $\psi_1(x) = \psi_2(x)$  for all  $x \in \mathcal{M}^3$ .

*Proof of Lemma.* Consider the functional  $\mathcal{I}$  that measures the discrepancy between  $\psi_1(x)$  and  $\psi_2(x)$ .

$$\begin{aligned} \mathcal{I}(\psi_1, \psi_2)(x) &:= \int_0^1 \left\{ \frac{d}{d\nu} [\nabla^2(\nu\psi_1(x) + (1-\nu)\psi_2(x)) - f(x, \nu\psi_1(x) + (1-\nu)\psi_2(x))] \right\} d\nu \\ &= [\nabla^2\psi_1 - f(x, \psi_1)] - [\nabla^2\psi_2 - f(x, \psi_2)] \end{aligned} \quad (33)$$

by the fundamental theorem of calculus. So  $\mathcal{I} = 0$  for  $\psi_1$  and  $\psi_2$  to be solutions to equation (32). Solving the function now by considering  $\partial/\partial\nu$  yields

$$\begin{aligned} \mathcal{I}(\psi_1, \psi_2)(x) &= \int_0^1 \left\{ \nabla^2\psi_1 - \nabla^2\psi_2 - D^2 f(x, \nu\psi_1(x) + (1-\nu)\psi_2(x))(\psi_1 - \psi_2) \right\} d\nu \\ &= \nabla^2(\psi_1 - \psi_2) - \left[ \int_0^1 D^2 f(x, \nu\psi_1 + (1-\nu)\psi_2) d\nu \right] (\psi_1 - \psi_2) \end{aligned} \quad (34)$$

where  $D^2$  is an operator which takes the partial derivative of  $f$  with respect to its second input. By the same logic as before, we require

$$\nabla^2(\psi_1 - \psi_2) = \mathcal{H}[x, \psi_1, \psi_2](\psi_1 - \psi_2) \quad (35)$$

where  $\mathcal{H}[x, \psi_1, \psi_2] := \int_0^1 D^2 f(x, \nu\psi_1 + (1-\nu)\psi_2) d\nu$ .

Specifically on the manifold, if we multiply both sides of equation (35) by  $(\psi_1 - \psi_2)$  and integrate,

$$\int_{\mathcal{M}^3} \mathcal{H} |\psi_1 - \psi_2|^2 = \int_{\mathcal{M}^3} |\nabla(\psi_1 - \psi_2)|^2 = 0 \quad (36)$$

and so if  $\mathcal{H} \geq 0$  and not identically zero, then equation (34) only holds if and only if  $\psi_1 - \psi_2 = 0$ . Also given the conditions stated in equation (31), then we must have that  $\psi_1 - \psi_2 = 0$  for  $\mathcal{I}$  to be zero. It is then only necessary that  $\psi_1(x)$  and  $\psi_2(x)$  take values in the interval  $I \subset \mathbb{R}$ .  $\square$

*Class*  $(\mathcal{Y}^-(\mathcal{M}^3), \sigma^2 \not\equiv 0, \tau \neq 0)$ . — Substituting for this class's initial data, the Lichnerowicz equation takes the form

$$\nabla_{(\lambda)}^2 \phi = f(x, \phi) = -\frac{1}{8}\phi - \frac{1}{8}\sigma^2\phi^{-7} + \frac{1}{12}\tau^2\phi^5 \quad (37)$$

and from the lemma we have

$$D^2 f(x, \phi) = \frac{\partial}{\partial \phi} f(x, \phi) = -\frac{1}{8} + \frac{7}{8}\sigma^2\phi^{-8} + \frac{5}{12}\tau^2\phi^4 \quad (38)$$

which needs to be non-negative but this is not necessarily clear from equation (38). Considering the Lichnerowicz equation for the interval  $I$  in which  $x_{min}$  of  $\phi$  yields

$$\begin{aligned} 0 &\leq \nabla_{(\lambda)}^2 \phi(x_{min}) = -\frac{1}{8}\phi(x_{min}) - \frac{1}{8}\sigma^2\phi(x_{min})^{-7} + \frac{1}{12}\tau^2\phi(x_{min})^5 \\ \implies \phi^4(x_{min}) &\geq \sqrt[4]{\frac{3}{2\tau^2}} \end{aligned} \quad (39)$$

as a lower bound for  $\phi$  regardless of the values of  $\tau$  and  $\sigma^2(x)$ . This restricts the interval  $I$  to

$$I = \left( \sqrt[4]{\frac{3}{2\tau^2}}, +\infty \right)$$

and hence,

$$\begin{aligned} \frac{\partial}{\partial \phi} f(x, \phi) &\geq \frac{\partial}{\partial \phi} f(x, \phi_{min}) \\ &= \frac{1}{2} + \frac{7}{18}\sigma^2\tau^4 \end{aligned} \quad (40)$$

which is clearly positive and therefore the solution is unique by the lemma.

*Class*  $(\mathcal{Y}^0(\mathcal{M}^3), \sigma^2 \not\equiv 0, \tau \neq 0)$ . — Again with  $R_{(\lambda)} = 0$  the Lichnerowicz equation takes the form

$$\nabla_{(\lambda)}^2 \phi = f(x, \phi) = -\frac{1}{8}\sigma^2\phi^{-7} + \frac{1}{12}\tau^2\phi^5 \quad (41)$$

$$\implies \frac{\partial}{\partial \phi} f(x, \phi) = \frac{7}{8}\sigma^2\phi^{-8} + \frac{5}{12}\tau^2\phi^4 \quad (42)$$

Given the even powers of  $\phi$ ,  $\frac{\partial}{\partial \phi} f(x, \phi) > 0$  and the solution is unique by the lemma.

For any other uniqueness solution the same process is repeated. The first derivatives with respect to  $\phi$  is found if possible, and the solution is computed. Otherwise,  $x_{min}$  and  $x_{max}$  are considered on the interval  $I$ .

#### 4. Further Discussion & Conclusion

Much of the discussion is embedded in the insights gained from the proofs to the existence and uniqueness of solutions. However, we add some further comments to our approach.

##### 4.1. Overview.

In this paper, we aimed to investigate a specific class of the Einstein Constraint Equations on a closed manifold under suitable assumptions. This is heavily motivated by the fact that the general solution to the Einstein Field Equation does not clearly exist and requires underlying simplifications. These included starting from the vacuum and  $\Lambda = 0$  version of the EFE and finding solutions for the parameters  $(g, K)$  from the initial data  $(\lambda, \sigma, \tau; \phi)$  (see equations (6a-b)). A crucial simplification to the Lichnerowicz equation was possible due to a conformal formulation of the metric and the existence of constant mean curvature  $\tau = \text{const}$  which led to the decoupling of the elliptic system of equations. The results from Yamabe Classification led to a simplification in the choice of scalar curvature, and importantly, the

sub/super-solution method allowed for generating explicit numerical solutions as seen in Table 1. In summary, the results include deriving the semi-decoupled PDEs

$$\nabla_{(\lambda)}^2 \phi = \frac{1}{8} R \phi - \frac{1}{8} (\sigma^{ij} + L W^{ij}) (\sigma_{ij} + L W_{ij}) \sigma^{-7} + \frac{1}{12} \tau^2 \phi^5 \quad (43)$$

$$\nabla_{(\lambda)}^i (L W)_m^i = \frac{2}{3} \phi^6 \nabla_m \tau \quad (44)$$

using the sequence  $\{(\sigma, W)\}$  and showing that this sequence converges to a solution to these equations via verifying the existence of  $\{(\phi, W)\}$ .

It is very clear that progressions in finding generalised solutions to the ECE is highly dependent on the underlying assumptions and must be investigated further in the non-CMC cases. Advancement in this field of study is still fairly recent and with more up and coming strategies and overarching surveys around the topic, finding generalisations for the non-CMC case is highly anticipated.

#### 4.2. Far-from-CMC.

The “closeness” to a constant mean curvature (CMC) solution is determined by the magnitude of  $\nabla \tau / \tau$  where we recall  $\tau = \text{tr}_g K$ . Using the methods stated in this paper, near-CMC solutions (constant or nearly constant  $\tau$ ) are possible with suitable assumptions. Removing the restriction on constant  $\tau$  leads to a much more difficult problem and therefore further results are limited by  $\nabla \tau$ . A quintessential example of solutions in *far-from-CMC regime* was suggested by Holst et al. [8] in which  $\sigma$  is suitably small but no further CMC assumptions are required. Substantially, the results were achieved due to a new topological fixed-point argument (see [7]), as well as a stronger notion of global sub/super-solutions method which we briefly mention here (see [3]).

In a similar vein to the Lichenrowicz equation (7) and the conformal killing factor  $L$  and vector field  $W$ , define

$$T(\phi, W) := \nabla_{(\lambda)}^2 \phi - \frac{1}{8} R \phi + \frac{1}{8} \|\sigma + L_\lambda W\|^2 \phi^{-7} - \frac{1}{12} \tau^2 \phi^5. \quad (45)$$

Then for positive functions  $\phi^\pm \in C^{2,\alpha}(M)$  are respectively *global* super/sub solutions if

$$T(\phi^+, W_\lambda) \leq 0 \quad \text{such that } (0 \leq \phi \leq \phi^+) \quad (46a)$$

$$T(\phi^-, W_\lambda) \geq 0 \quad \text{such that } (\phi^- \leq \phi) \quad (46b)$$

for all  $\phi \in C^{2,\alpha}(M)$ . According to Holst et al. [8], given any generalised initial data  $(\lambda, \sigma, \tau)$ , assume the existence of a global sub-solution  $\phi^-$  and super-solution  $\phi^+$  such that  $\phi^- \leq \phi^+$ . Then there exist a solution  $(\phi, W)$  for the generalised vacuum, elliptic system similar to equations (6a-b).

### 5. Future Work

It is clear that the results shown are restricted to the near-CMC case within vacuum space-time. Although major progress since the publication of [9] has been accomplished, some of the following open questions (see [3]) remain unsolved and suitable for further research.

- *Conformal method as generic parametrisation of initial data*
  - There are no unique solutions under the conformal data and the near-CMC and transverse-traceless tensor (TT-hypothesis) assumptions [10]. Given the negative results, is it possible to formulate another suitable parametrisation with a more generalised approach.
- *CMC metrics as differentiable manifold*

- Only the vacuum Einstein Field Equations ( $\Lambda = 0$ ) were considered. It is known that the class of scalar-flat Riemannian metrics is a closed differentiable manifold and leads to similar results [5]. In that case, is it possible to for a class or metrics with  $\tau = 6$  (derived when  $\Lambda > 0$ ) to be a closed differentiable manifold?
- *Massive data that are flat in arbitrary domains*
  - Can one construct asymptotically flat, scalar-flat metrics on  $\mathbb{R}^3$  that are generalised to exist on an arbitrary regular region  $\Omega$  under the assumption that  $\mathbb{R}^3 \setminus \Omega$  is not bounded and does not lead to vanishing mass?

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